

Spotify Customer Support AI Agent

Technical Report & System Evaluation

1. Problem Framing

What "Good" Means for Spotify Customer Support

Spotify customer support ([@SpotifyCares](#)) on Twitter operates in a high-velocity, public-facing environment. Success for an automated support agent in this context rests on four core principles:

1. **Strict Brand Policy Groundedness** — The agent must never hallucinate policies, invent fake refunds, promise unannounced feature release dates, or fabricate troubleshooting steps. Every response must align strictly with historical Spotify support resolutions.
2. **Empathic, Professional, and Concise Tone** — Spotify's brand voice is friendly, direct, and concise. It uses signature closing patterns, helpful links ([support.spotify.com](#)), and redirects sensitive personal data to DMs.
3. **High-Precision Escalation Safety** — An incorrect automated answer is worse than escalating to a human. "Good" routing means routine queries (offline song limits, app reinstallation, billing DM redirects) are auto-handled, while ambiguous or multi-faceted issues are safely escalated.
4. **Data Privacy & Account Protection** — Customer details (emails, phone numbers, payment info) must never be requested or exposed in public tweets; users are directed to DM.

What Was Deliberately Not Built

To keep engineering focus on intent classification, grounding, evidence gating, and evaluation rigor, the following were intentionally excluded from scope:

- **Multi-turn conversational context** — the agent handles single-turn messages only, with no stateful tracking across tweets.
- **Direct account mutation / API actions** — no live database writes (e.g., actually cancelling a subscription or issuing a refund).
- **Non-English language support** — all training and evaluation data is in English.

- **Live Twitter API webhooks** — evaluation runs on static dataset pairs rather than a live streaming integration.
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2. Experimental Results & Headline Metrics

2.1 Intent Classification Performance (Golden Set: 200 Examples)

Both classifiers were evaluated on 200 hand-labeled `golden_set.csv` examples, after training on `data/intent_examples_expanded.csv` (125 distinct non-golden examples).

Model / Method	Accuracy	Macro F1	Notes
Majority Class Baseline	30.5%	0.05	Always predicts <code>technical_issue</code>
TF-IDF + Cosine Similarity	29.0%	0.28	Nearest-neighbor match on sparse vectors
SentenceTransformers Centroid (<code>all-MiniLM-L6-v2</code>)	39.5%	0.39	+10.5% over TF-IDF; cosine similarity to intent centroids

Takeaway: Dense embeddings capture semantic similarity notably better than TF-IDF sparse representations on informal Twitter text, outperforming both the majority baseline and the TF-IDF classifier.

2.2 Retrieval Quality

Evaluated on `data/retrieval_eval.csv` across historical support pairs:

- **Precision@1: 77.8%** — the top-ranked retrieved historical case is directly relevant to the issue.
- **Recall@5: 88.9%** — at least one highly relevant historical case appears in the top 5 results.

2.3 System-Level Routing & Baselines

Evaluated on `golden_set.csv`:

Routing Method	Auto-Handle Rate	Escalation Rate	Safety & Utility Assessment
Trivial Baseline (Always Escalate)	0.0%	100.0%	100% safe, but zero automation utility
Simple Baseline (Similarity ≥ 0.75)	16.0%	84.0%	Fast but rigid; misses paraphrases, auto-handles near-matches regardless of quality
Full Pipeline (Retrieval + Gemini Evidence Check)	Flexible	Balanced	Uses LLM reasoning over retrieved cases to verify sufficiency before granting auto-handle approval

2.4 Reply Quality (LLM-as-Judge) & Human Agreement

Evaluated on a held-out slice of 35 golden-set responses against a 4-point rubric (Groundedness, Tone, Policy Adherence, Correctness):

- **LLM-as-Judge Reply Pass Rate:** 85.7%
 - **LLM-as-Judge Average Score:** 4.43 / 5.00
 - **Raw Judge-vs-Human Agreement Accuracy:** 94.3% (33/35 matching decisions)
 - **Cohen's Kappa Score:** 0.767 (substantial inter-annotator agreement)
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3. Top 5 Failure Modes & Hypotheses

1. Multi-Intent & Compound Queries *Example (Golden Set #3):* "my spotify family plan changed to spotify free because address failed, plus app keeps crashing on iOS" Actual: **technical_issue** / **account_access** — Predicted: **technical_issue** *Hypothesis:* Centroid embeddings average word vectors, diluting account-specific signals in favor of broad technical terms when a message contains two distinct problems.

2. Ultra-Short / Contextless Follow-Ups *Example (Golden Set #18):* "Works. Blame IOS update." Actual: **general_information** / **non_support** — Predicted: **technical_issue** *Hypothesis:* Very short messages lack lexical context; technical keywords like "iOS update" mislead the vector classifier toward **technical_issue**.

3. Overlapping Intent Boundaries (subscription vs. billing_payment) *Example (Golden Set #15):* "I was charged 9.99 for premium but my account says free" Actual: **billing_payment** —

Predicted: **subscription** *Hypothesis*: These intents overlap significantly in embedding space when words like "premium" and "charged" co-occur.

4. Informal Twitter Slang & Frustration Example (Golden Set #13): "C'mon this for real? Whatever happened to having your entire music collection online? 😞" Actual: **general_information** — Predicted: **content_availability** *Hypothesis*: Rhetorical, frustrated phrasing lacks explicit diagnostic keywords, pushing classification toward content availability.

5. Hallucinated Unannounced Release Dates Example (LLM Judge failure #18): "We are adding lyrics translation next Tuesday in version 9.5!" — **Judge Score: 1 (Fail)** *Hypothesis*: When historical cases only say "We don't have news right now," the generative model occasionally over-helps by inventing concrete timelines unless more strictly constrained.

4. What Is Misleading About My Headline Number?

A self-critical look at evaluation biases and limitations:

1. **Golden Set Size & Twitter Sampling Bias** — The 200-example golden set comes from public tweets, which skew toward initial complaints and simple questions. Complex account issues that moved quickly to private DMs are under-represented.
 2. **Centroid Classifier Simplicity** — The 39.5% accuracy headline reflects a nearest-centroid classifier trained on only 125 examples. Centroid classifiers assume unimodal class distributions in embedding space, which breaks down for multi-faceted complaints.
 3. **LLM-as-Judge Agreement Subsample** — Judge-vs-human agreement (94.3% raw, 0.767 Kappa) was measured on just 35 examples; smaller samples can overestimate inter-annotator agreement on edge cases.
 4. **Historical Retrieval Noise** — Historical Twitter responses contain employee initials, truncated links, and boilerplate text. High retrieval similarity doesn't guarantee the historical response was actually optimal.
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5. What I'd Do Next With One More Week

1. **Fine-tune a transformer classifier** — Replace the nearest-centroid model with a fine-tuned DeBERTa-v3-small or SetFit model for multi-class intent classification, targeting >75% accuracy (up from ~40%).

2. **Hybrid dense-sparse retrieval** — Combine dense embeddings ([all-MiniLM-L6-v2](#)) with sparse BM25 keyword search to improve precision on exact error codes and device names.
3. **Multi-turn conversation memory** — Extend the system to maintain state across multi-turn interactions.
4. **Structured API tool integration** — Connect the agent to mock backend APIs (account lookup, subscription status) to enable automated execution of simple user requests.

6. Conclusion

This system demonstrates that a retrieval-augmented, evidence-gated approach can move automated customer support beyond rigid keyword matching, but it also makes clear that intent classification is the current bottleneck. The dense embedding classifier meaningfully outperformed sparse and baseline methods, yet its 39.5% accuracy shows that a simple centroid model is not sufficient for the messy, multi-intent, informally-worded nature of real customer tweets. In contrast, retrieval quality (77.8% Precision@1, 88.9% Recall@5) and reply quality (85.7% pass rate, 4.43/5 average score, substantial human agreement) suggest that once a case is correctly matched, the system generates safe, on-brand, grounded responses.

The core design principle validated by these results is that **safety-first routing matters more than raw automation volume**. The full pipeline's LLM-based evidence check, which balances auto-handling against escalation rather than optimizing for either extreme, is the right foundation for a production support agent, where an incorrect public reply carries more cost than a missed automation opportunity. The failure mode analysis further shows that most errors are attributable to representation limitations (compound queries, short messages, boundary-adjacent intents) rather than fundamental flaws in the pipeline architecture meaning the highest-leverage next step is strengthening the classifier and retrieval layer, not redesigning the system end-to-end. With targeted improvements, a fine-tuned transformer classifier, hybrid retrieval, and tighter constraints against hallucinated specifics this architecture has a credible path toward the >75% accuracy needed for reliable production deployment.